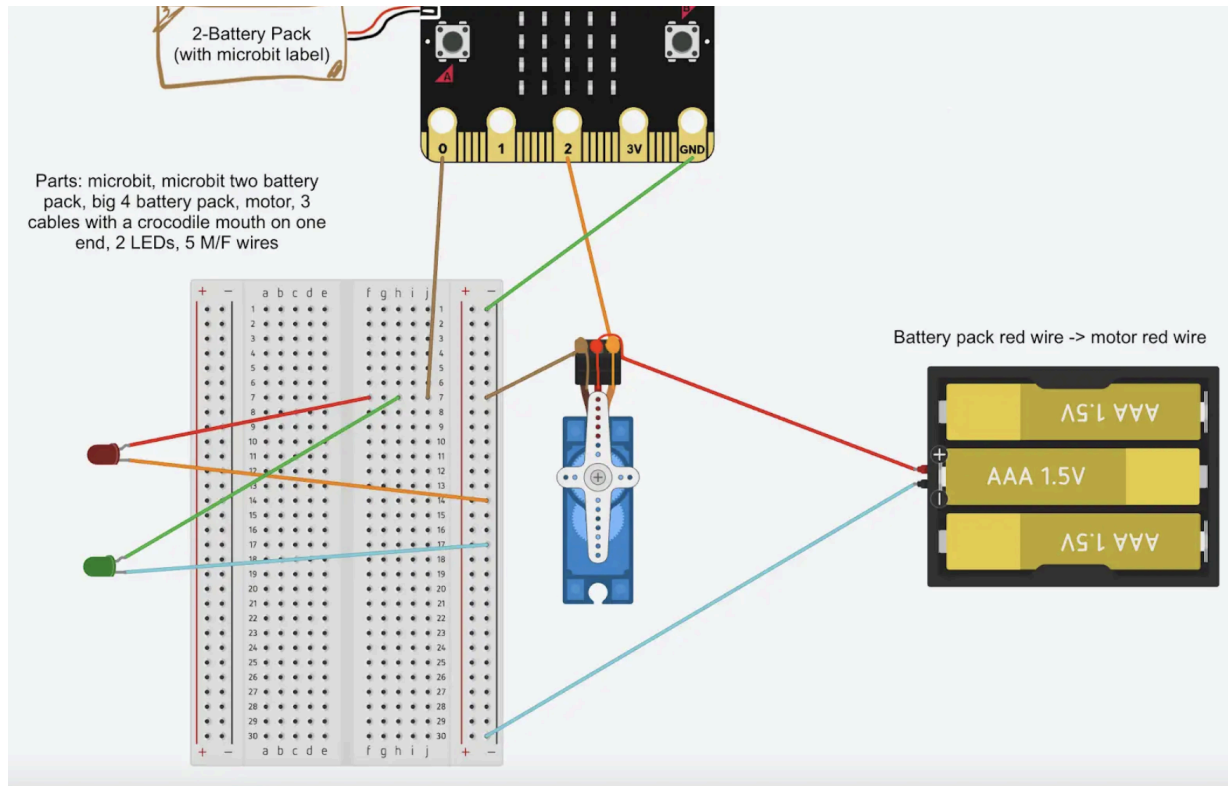




Steps with Breadboard

Parts Checklist

- ☐ Microbit
- ☐ Breadboard
- ☐ Big Battery Pack
- ☐ Small Battery Pack
- ☐ 4 Wires Alligator-Pin
- ☐ Motor
- ☐ 2 LEDs
- ☐ Breadboard
- ☐ 4 Pin/Hole Wires
 - ☐ 4 for LEDs to be attached to breadboard
 - ☐ 1 for Motor attached to battery pack
- ☐ 2 Pin/Pin Wires
 - ☐ 2 for Motor attached to breadboard

1. LEDs

- a. Take both LEDs and 4 Pin/Hole wires and attach all F ends to LED pins
 - i. Make note of which wires are the longer/shorter legs of LED

1. Ex: Yellow is the short leg, Green is the long leg
2. What does this mean?
 - a. Short leg (-) is goes to GND, Longer leg (+) goes to 3V
 - ii. Take the wire attached to the shorter leg, and plug it into the negative column on the breadboard, in any row - do this for both LEDs
- b. Connect LED 1
 - i. Take Alligator Clip 1, and clamp it to Pin 1 on Microbit
 - ii. With other end of wire, plug it in to Row 1 on the breadboard
 - iii. Take LED 1, and plug the remaining wire (which should be connected to the longer leg) and plug it into Row 1
- c. Connect LED 2
 - i. Take Alligator Clip 2, and clamp it to Pin 2 on Microbit
 - ii. With other end of wire, plug it in to Row 2 on the breadboard
 - iii. Take LED 2, and plug its remaining wire (which should be connected to the longer leg) and plug it into Row 2
2. Motor
 - a. Take 2 Pin/Pin wires, and 1 Pin/Hole wire and attach to the grouped wires coming from the motor
 - i. Keep track of which wire corresponds to each color of the grouped wires
 - ii. The Pin/Hole wire should correspond to the RED wire coming from the motor
 - b. With the wire corresponding to the BLACK wire coming from the motor, plug it into the negative column on the breadboard
 - c. With the wire corresponding to the YELLOW wire coming from the motor, plug it into Row 3 on the breadboard
 - d. Take Alligator Clip 3, and clamp it on to Pin 0 on Microbit
 - e. With other end of the wire, plug it in to Row 3 on the breadboard
3. GND
 - a. Take Alligator Clip 4, and clamp it to GND on Microbit
 - b. With other end of wire, plug in to negative column on breadboard in any row
4. Batteries
 - a. Plug small battery back into microbit
 - b. Take the Pin/Hole wire attached to the red wire coming from the motor, and attach the F end to the Red wire coming from the big battery pack
 - c. Take the back wire coming from the big battery pack and plug it into the negative column on the breadboard in any row
 - d. Turn on the big battery pack

Steps without Full Breadboard

Parts Checklist

- ☐ Microbit
- ☐ Negative/Positive Column of the Breadboard
- ☐ Big Battery Pack

- ☐ Small Battery Pack
- ☐ 4 Wires Alligator-Pin
- ☐ Motor
- ☐ 2 LEDs
- ☐ Breadboard
- ☐ 5 Pin/Hole Wires
 - ☐ 4 for LEDs to be attached to breadboard
 - ☐ 1 for Motor attached to battery pack
- ☐ 1 Pin/Pin Wires
 - ☐ 1 for Motor attached to breadboard neg column
- ☐ 3+ Hole/Hole Wires
 - ☐ 2 for LED-MB
 - ☐ 1 for Motor-MB

1. LEDs

- a. Take both LEDs and 4 Pin/Hole wires and attach all F ends to LED pins
 - i. Make note of which wires are the longer/shorter legs of LED
 1. Ex: Yellow is the short leg, Green is the long leg
 2. What does this mean?
 - a. Short leg (-) is goes to GND, Longer leg (+) goes to 3V
 - ii. Take the wire attached to the shorter leg, and plug it into the negative column on the breadboard, in any row - do this for both LEDs
- b. Connect LED 1
 - i. Take Alligator Clip 1, and clamp it to Pin 1 on Microbit
 - ii. With other end of wire, plug it in to Hole/Hole wire
 - iii. Take LED 1, and plug the remaining wire (which should be connected to the longer leg) and plug it into the other end of the Hole/Hole wire
- c. Connect LED 2
 - i. Take Alligator Clip 2, and clamp it to Pin 2 on Microbit
 - ii. With other end of wire, plug it in to Hole/Hole wire
 - iii. Take LED 2, and plug its remaining wire (which should be connected to the longer leg) and plug it into the other end of the Hole/Hole wire

2. Motor

- a. Take 1 Pin/Pin wire, 1 Alligator Clip-M wire, and 1 Pin/Hole wire and attach to the grouped wires coming from the motor
 - i. Keep track of which wire corresponds to each color of the grouped wires
 - ii. The Pin/Hole wire should correspond to the RED wire coming from the motor
 - iii. The Pin/Pin wire should correspond with the BLACK wire
 - iv. The Alligator Pin should correspond with the YELLOW wire
- b. With the wire corresponding to the BLACK wire coming from the motor, plug it into the negative column on the breadboard

- c. With the wire corresponding to the YELLOW wire coming from the motor, clamp it on to Pin 0 of the microbit
 - d. Take Alligator Clip 3, and clamp it on to Pin 0 on Microbit
- 3. GND
 - a. Take Alligator Clip 4, and clamp it to GND on Microbit
 - b. With other end of wire, plug in to negative column on breadboard in any row
- 4. Batteries
 - a. Plug small battery back into microbit
 - b. Take the Pin/Hole wire attached to the red wire coming from the motor, and attach the F end to the Red wire coming from the big battery pack
 - c. Take the back wire coming from the big battery pack and plug it into the negative column on the breadboard in any row
 - d. Turn on the big battery pack

Steps with Ceibal's Board

Parts Checklist

- ☐ Microbit
 - ☐ Ceibal Board -
 - ☐ Big Battery Pack
 - ☐ Small Battery Pack
 - ☐ 4 Wires Alligator-Pin
 - ☐ Motor
 - ☐ 2 LEDs
 - ☐ Breadboard
 - ☐ 4 Pin/Hole Wires
 - ☐ 4 for LEDs to be attached to breadboard
 - ☐ 1 for Motor attached to battery pack
 - ☐ 2 Pin/Pin Wires
 - ☐ 2 for Motor attached to breadboard
-
- 5. LEDs
 - a. Take both LEDs and 4 Pin/Hole wires and attach all F ends to LED pins
 - i. Make note of which wires are the longer/shorter legs of LED
 - 1. Ex: Yellow is the short leg, Green is the long leg
 - 2. What does this mean?
 - a. Short leg (-) is goes to GND, Longer leg (+) goes to 3V
 - ii. Take the wire attached to the shorter leg, and plug it into row 10 of the board (just has to be same as wherever MB's ground is) - do this for both LEDs
 - b. Connect LED 1
 - i. Take Alligator Clip 1, and clamp it to Pin 1 on Microbit
 - ii. With other end of wire, plug it in to Row 1 on the breadboard

- iii. Take LED 1, and plug the remaining wire (which should be connected to the longer leg) and plug it into Row 1
 - c. Connect LED 2
 - i. Take Alligator Clip 2, and clamp it to Pin 2 on Microbit
 - ii. With other end of wire, plug it in to Row 2 on the breadboard
 - iii. Take LED 2, and plug its remaining wire (which should be connected to the longer leg) and plug it into Row 2
- 6. Motor
 - a. Take 2 Pin/Pin wires, and 1 Pin/Hole wire and attach to the grouped wires coming from the motor
 - i. Keep track of which wire corresponds to each color of the grouped wires
 - ii. The Pin/Hole wire should correspond to the RED wire coming from the motor
 - b. With the wire corresponding to the BLACK wire coming from the motor, plug it into row 10 of the board (just has to be same as wherever MB's ground is)
 - c. With the wire corresponding to the YELLOW wire coming from the motor, plug it into Row 3 on the breadboard
 - d. Take Alligator Clip 3, and clamp it on to Pin 0 on Microbit
 - e. With other end of the wire, plug it in to Row 3 on the breadboard
- 7. GND
 - a. Take Alligator Clip 4, and clamp it to GND on Microbit
 - b. With other end of wire, plug in to row 10 of the board
- 8. Batteries
 - a. Plug small battery back into microbit
 - b. Take the Pin/Hole wire attached to the red wire coming from the motor, and attach the F end to the Red wire coming from the big battery pack
 - c. Take the back wire coming from the big battery pack and plug it into the negative column on the breadboard in any row
 - d. Turn on the big battery pack

Converting to ScratchJr Project

Steps with Breadboard

Parts Checklist

- ☐ Microbit
- ☐ Breadboard
- ☐ Big Battery Pack
- ☐ Small Battery Pack
- ☐ 4 Wires Alligator-Pin
- ☐ Motor
- ☐ 2 LEDs
- ☐ 4 Pin/Hole Wires
 - ☐ 4 for LEDs to be attached to breadboard
 - ☐ 1 for Motor attached to battery pack
- ☐ 2 Pin/Pin Wires
 - ☐ 2 for Motor attached to breadboard

9. LEDs

- a. Take both LEDs and 4 Pin/Hole wires and attach all F ends to LED pins
 - i. Make note of which wires are the longer/shorter legs of LED
 - 1. Ex: Yellow is the short leg, Green is the long leg
 - 2. What does this mean?
 - a. Short leg (-) is goes to GND, Longer leg (+) goes to 3V
 - ii. Take the wire attached to the shorter leg, and plug it into the negative column on the breadboard, in any row - do this for both LEDs
- b. Connect LED 1
 - i. Take Alligator Clip 1, and clamp it to Pin 1 on Microbit
 - ii. With other end of wire, plug it in to Row 1 on the breadboard
 - iii. Take LED 1, and plug the remaining wire (which should be connected to the longer leg) and plug it into Row 1
- c. Connect LED 2
 - i. Take Alligator Clip 2, and clamp it to Pin 2 on Microbit
 - ii. With other end of wire, plug it in to Row 2 on the breadboard
 - iii. Take LED 2, and plug its remaining wire (which should be connected to the longer leg) and plug it into Row 2

10. Motor

- a. Take 2 Pin/Pin wires, and 1 Pin/Hole wire and attach to the grouped wires coming from the motor
 - i. Keep track of which wire corresponds to each color of the grouped wires
 - ii. The Pin/Hole wire should correspond to the RED wire coming from the motor
- b. With the wire corresponding to the BLACK wire coming from the motor, plug it into the negative column on the breadboard

- c. With the wire corresponding to the YELLOW wire coming from the motor, plug it into Row 3 on the breadboard
- d. Take Alligator Clip 3, and clamp it on to Pin 0 on Microbit
- e. With other end of the wire, plug it in to Row 3 on the breadboard

11. GND

- a. Take Alligator Clip 4, and clamp it to GND on Microbit
- b. With other end of wire, plug in to negative column on breadboard in any row

12. Batteries

- a. Plug small battery back into microbit
- b. Take the Pin/Hole wire attached to the red wire coming from the motor, and attach the F end to the Red wire coming from the big battery pack
- c. Take the back wire coming from the big battery pack and plug it into the negative column on the breadboard in any row
- d. Turn on the big battery pack